

Decision and Social Choice

PE3101P — Lecture 8: Egalitarianism and prioritarianism

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8th October, 2026

Today

- 1 More on Harsanyi
- 2 Different Numbers of People
- 3 Egalitarianism
- 4 Prioritarianism
- 5 Ex Ante and Ex Post
- 6 Priority and Risk Aversion

Harsanyi's Aggregation Theorem

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Utilitarianism

The best outcome is the one with the greatest sum of wellbeing*:

$$X \succsim Y \quad \text{iff} \quad w_1(X) + \cdots + w_n(X) \geq w_1(Y) + \cdots + w_n(Y),$$

where $w_i(X)$ is person i 's wellbeing in outcome X .

*Strictly, a sum of utilities linked to wellbeing: VNM utilitarianism.

Decision Theory and Social Choice

Decision theory

	Heads	Tails
Option A	5	5
Option B	11	0

Utilities, fair coin.

Social choice

	Alex	Darius
Option A	5	5
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Wellbeing levels.

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- ▶ Decision theory deals with risk: an option can be better in some states of nature and worse in others.

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Wellbeing levels.

- ▶ Decision theory deals with risk: an option can be better in some states of nature and worse in others.
- ▶ Social choice deals with trade-offs between people: an option can be better for some people and worse for others.
- ▶ States of nature play the role that people play. Statewise Dominance $\leftrightarrow$ Pareto; risk $\leftrightarrow$ inequality.

Decision Theory and Social Choice



A visual representation of the differences between decision theory and social choice theory.

Two Analogous Principles

Anonymity

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Stochastic Equivalence

If two gambles give the same outcomes with the same probabilities, just in different states of nature, they are equally good.

- ▶ Structurally, these are the same principle: one rearranges people, the other rearranges states.
- ▶ That doesn't make them equally plausible.

A Two-Way Connection

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- ▶ We can still reduce the value of an outcome or gamble to how good its lottery-equalisation is for an individual: an equal chance of being each person.
- ▶ E.g. $(10, 0) \sim (5, 5)$ iff a 50–50 chance of 10 or 0 is as good for a person as 5 for sure.

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- ▶ E.g. $(10, 0) \sim (5, 5)$ iff a 50–50 chance of 10 or 0 is as good for a person as 5 for sure.
- ▶ Interestingly, Anonymity for outcomes then corresponds to Stochastic Equivalence for gambles over equally likely states.

A Simple Case (1): UNEQUAL and EQUAL CHANCES

UNEQUAL

	Heads	Tails
Alex	10	10
Darius	0	0

UNEQUAL for sure, written out with a fair coin:
Alex gets 10 and Darius gets 0, however it lands.

A Simple Case (1): UNEQUAL and EQUAL CHANCES

	Heads	Tails
Alex	10	10
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Now look at each state of nature separately.
If Tails, the outcome is (10, 0).

A Simple Case (1): UNEQUAL and EQUAL CHANCES

	Heads	Tails
Alex	10	0
Darius	0	10

Swap Alex and Darius in Tails: (0, 10).
By Anonymity, this is equally good.

A Simple Case (1): UNEQUAL and EQUAL CHANCES

EQUAL CHANCES

	Heads	Tails
Alex	10	0
Darius	0	10

Equally good in every state (Heads is unchanged).

By Statewise Dominance, both ways round:

UNEQUAL $\sim$ EQUAL CHANCES.

From Pairwise Anonymity to Anonymity (1)

Pairwise Anonymity

If outcome Y can be obtained from outcome X by swapping the wellbeing levels of two people, then $X \sim Y$.

(1, 2, 3, 4, 5, 6)

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- ▶ So, by Transitivity, any rearrangement is equally good: Anonymity.
- ▶ This works for any finite number of people. With infinitely many, it doesn't.

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Permutation Invariance

For any rearrangement π : $X \succsim Y$ iff $\pi X \succsim \pi Y$.

- Permutation Invariance rearranges both outcomes in the same way. Swapping Alex and Darius: $(10, 0) \succsim (5, 5)$ iff $(0, 10) \succsim (5, 5)$.

How the Two Principles Differ

- ▶ Anonymity and Transitivity give Permutation Invariance: if $X \succsim Y$, then $\pi X \sim X \succsim Y \sim \pi Y$.

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 - ▶ That satisfies Permutation Invariance: rearranging both outcomes the same way keeps who is better off where.
 - ▶ But (10, 0) and (0, 10) are incomparable. Anonymity fails.
- ▶ What Permutation Invariance does rule out: a rearrangement being *better* than the original.

Permutation Invariance: No Rearrangement Is Better

Write $\pi^2 X$ for $\pi(\pi X)$: π applied twice. Likewise $\pi^3 X$, and so on.

Claim

Given Permutation Invariance and Transitivity, $\pi X \succ X$ is impossible.

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- ▶ For some n , $\pi^n X = X$. (Every rearrangement of finitely many people, repeated enough times, gets back to the start: the cycle decomposition theorem.)
- ▶ Transitivity: $X = \pi^n X \succ \pi^{n-1} X \succ \dots \succ \pi X \succ X$. So $X \succ X$: contradiction.

With Completeness: Anonymity

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- ▶ Since neither is better than the other, both hold: $X \sim \pi X$. That's Anonymity.
- ▶ So the two principles differ only where there is incomparability.

Equally Good, or Incomparable?

- ▶ Permutation Invariance allows two verdicts about $(10, 0)$ and $(0, 10)$:
 - ▶ equally good: that's Anonymity;
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- ▶ If they're equally good, then $(11, 0) \succ (10, 0) \sim (0, 10)$: Alex getting 11 is better than Darius getting 10.
- ▶ That's not obvious. Incomparability can survive small improvements: $(11, 0)$ might still be neither better nor worse than $(0, 10)$.

Keeping Permutation Invariance, Dropping Anonymity

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- ▶ So could EQUAL CHANCES be better than UNEQUAL , as the fairness intuition says?

Keeping Permutation Invariance, Dropping Anonymity

- ▶ In the simple case, Anonymity and Statewise Dominance gave $UNEQUAL \sim EQUAL \text{ CHANCES}$.
- ▶ If $(10, 0)$ and $(0, 10)$ are incomparable, Statewise Dominance no longer gives that.
- ▶ So could $EQUAL \text{ CHANCES}$ be better than $UNEQUAL$, as the fairness intuition says?
- ▶ It would be better than $(10, 0)$ for sure, even though in each state its outcome is $(10, 0)$ or $(0, 10)$, neither of which is better than $(10, 0)$.

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Negative Dominance

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- ▶ Many people find Negative Dominance quite plausible.

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- ▶ EQUAL CHANCES would be better than $(10, 0)$ for sure, though none of its outcomes is better than $(10, 0)$.
- ▶ Many people find Negative Dominance quite plausible.
- ▶ So perhaps one could avoid the theorem's conclusion by dropping Anonymity. But it's not so clear that that's a good way of doing it.

Ex Ante Pareto, One Person at a Time

EQUAL CHANCES		
	Heads	Tails
Alex	10	0
Darius	0	10

Last week's second move, again.

Ex Ante Pareto, One Person at a Time

	Heads	Tails
Alex	10	0
Darius	0	10

Alex's expected wellbeing: $\frac{1}{2} \cdot 10 + \frac{1}{2} \cdot 0 = 5$.

Ex Ante Pareto, One Person at a Time

	Heads	Tails
Alex	5	5
Darius	0	10

Give Alex 5, however the coin lands.
His expected wellbeing is still 5.

Ex Ante Pareto, One Person at a Time

	Heads	Tails
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Darius's expected wellbeing: $\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 10 = 5$.

Ex Ante Pareto, One Person at a Time

	Heads	Tails
Alex	5	5
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Give Darius 5, however the coin lands.
His expected wellbeing is still 5.

Ex Ante Pareto, One Person at a Time

EQUAL		
	Heads	Tails
Alex	5	5
Darius	5	5

By Expected Utility, equally good for each person.
By Ex Ante Pareto, both ways round:
EQUAL CHANCES $\sim$ EQUAL.

Ex Ante Pareto, One Person at a Time

EQUAL		
	Heads	Tails
Alex	5	5
Darius	5	5

Each step changed only one person's gamble. So, with Transitivity, we only need Ex Ante Pareto for changes to one person at a time.

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- ▶ Why: concentrate each outcome's wellbeing in one person, then remove the neutral lives. Next slide.

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- ▶ Why: concentrate each outcome's wellbeing in one person, then remove the neutral lives. Next slide.
- ▶ Zero: 0 is the level where a life would be neither good nor bad. That fixes which lives are neutral.

Concentrate, Then Remove the Neutral Lives

A dash: that person doesn't exist.

	Alex	Darius	Philip	Total	Why
X	3	1	2	6	

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~	6	0	0	6	

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Y	2	5	—	7	

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Y	2	5	—	7	
~	0	7	—	7	The theorem
~	—	7	—	7	Neutral Addition
~	7	—	—	7	Anonymity across different people

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X	3	1	2	6	
$\sim$	6	0	0	6	The theorem
$\sim$	6	—	—	6	Neutral Addition
Y	2	5	—	7	
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$\sim$	—	7	—	7	Neutral Addition
$\sim$	7	—	—	7	Anonymity across different people

$(7, -, -) \succ (6, -, -)$: Alex at 7 is better than Alex at 6.

So, by Transitivity, $Y \succ X$: the greater total wins.

Egalitarianism

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Egalitarianism (telic)

It is in itself bad if some people are worse off than others.

- ▶ In principle, our social ordering $\succsim$ could be about either: which outcome *ought to be chosen*, or which outcome is *better*.
- ▶ I'll mostly have the betterness interpretation in mind.

Egalitarianism at Work

	Alex	Darius	Total
UNEQUAL	10	0	10
EQUAL	5	5	10

Egalitarianism at Work

	Alex	Darius	Total
UNEQUAL	10	0	10
EQUAL	5	5	10

- ▶ Utilitarianism: equally good. Same total.
- ▶ Telic egalitarianism: UNEQUAL is worse. Darius is worse off than Alex, and that is bad in itself.

Egalitarianism at Work

	Alex	Darius	Total
UNEQUAL	10	0	10
EQUAL	5	5	10

- ▶ Utilitarianism: equally good. Same total.
- ▶ Telic egalitarianism: UNEQUAL is worse. Darius is worse off than Alex, and that is bad in itself.
- ▶ Deontic egalitarianism: UNEQUAL should not be chosen.

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- ▶ If inequality is in itself bad, it is bad even when it has no bad effects.

The Divided World

- ▶ The two halves of the world's population don't know about each other.

	One half	Other half	Average
(1)	100	200	150
(2)	145	145	145

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- ▶ (1) has more wellbeing. Many people think (2) is better.
- ▶ The halves are unconnected, so the inequality in (1) has no effects. If (2) is better, inequality must be bad in itself: the telic view.

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- ▶ For Parfit, cases like these most clearly separate the two views.

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- ▶ The Levelling Down Objection: there is nothing good about it at all. It's not just that levelling down would be wrong: it isn't good in any way.
- ▶ Deontic egalitarianism avoids the objection: it needn't say the change is good in any way.

The Priority View

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- ▶ For a utilitarian, how much a benefit matters depends only on its size.
- ▶ For a prioritarian, it also depends on how well off the person who gets it is.
- ▶ The priority isn't absolute: a big enough benefit to someone better off can outweigh a small benefit to someone worse off.

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“People at higher altitudes find it harder to breathe.”

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- ▶ Prioritarians care only about *absolute levels*: how well off each person is.
- ▶ Parfit's comparison: at high altitude it's harder to breathe. That's because of how high you are, not because others are lower down.
- ▶ One can be both. But a prioritarian as such doesn't think inequality is bad in itself.

Two Reasons to Prefer EQUAL

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- ▶ Prioritarians favour equality only as a by-product of giving priority to the worse off.
- ▶ A utilitarian can also prefer equal *resources*: extra resources do less for someone who already has a lot. But not equal wellbeing.

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- ▶ So prioritarianism avoids the Levelling Down Objection.
- ▶ And it can still say (2) is better in the Divided World: the better off lose more than the worse off gain, but the worse off's gains matter more.

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- ▶ Prioritarianism has it built in: how much a benefit to someone matters depends only on how well off they are, whatever anyone else has.
- ▶ Egalitarianism needn't satisfy it: whether a change reduces inequality can depend on where the unaffected people are. Some egalitarian views happen to satisfy it; others don't.

Separability over Persons: An Example

	Alex	Darius	Philip
X	4	36	49
Y	9	16	49
X'	4	36	1
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- ▶ Prioritarianism with $f = \sqrt{\cdot}$: X beats Y , and X' beats Y' . Philip's term cancels: $2 + 6 = 8$ against $3 + 4 = 7$.
- ▶ An egalitarian view could rank Y above X (Alex is the worst off) but X' above Y' (now Philip is).

A Representation Theorem for Prioritarianism

Adler and Holtug (2019). Fixed population. Suppose the ranking of outcomes satisfies:

- ▶ Strong Pareto, Pareto Indifference and Anonymity;
- ▶ Pigou–Dalton: a transfer from someone better off to someone worse off, which doesn't swap their ranks, is an improvement;
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Then

It ranks outcomes by $g(w_1) + \dots + g(w_n)$, for some strictly increasing, strictly concave g . And every such ranking satisfies all of the above.

Writing Prioritarianism Down

Prioritarianism

$$X \succsim Y \quad \text{iff} \quad \sum_i f(w_i(X)) \geq \sum_i f(w_i(Y)),$$

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- ▶ For example, $f(w) = \sqrt{w}$: not the true one, just a convenient example for non-negative wellbeing.
 - ▶ (25, 25): $\sqrt{25} + \sqrt{25} = 5 + 5 = 10$.
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 - ▶ (25, 25): $\sqrt{25} + \sqrt{25} = 5 + 5 = 10$.
 - ▶ (49, 1): $\sqrt{49} + \sqrt{1} = 7 + 1 = 8$.
- ▶ So (25, 25) is better than (49, 1), though the totals are the same.

Prioritarianism and Utilitarianism Can Disagree

	Alex	Darius	Total	Priority-weighted total ($f(w) = \sqrt{w}$)
X	25	0	25	$5 + 0 = 5$
Y	9	9	18	$3 + 3 = 6$

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- ▶ Utilitarianism: X is better. It has the greater total.
- ▶ Prioritarianism with $f(w) = \sqrt{w}$: Y is better.
- ▶ Going from Y to X, Alex gains 16 and Darius loses 9.
Darius's loss counts for more, because he ends up so badly off.

Ex Ante and Ex Post Prioritarianism

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- ▶ The more common one is ex post prioritarianism.
- ▶ Square numbers from here: (49, 1) for sure; a coin toss giving (49, 1) or (1, 49); and (25, 25) for sure.

Ex Ante Prioritarianism at Work

Coin toss		
	Heads	Tails
Alex	49	1
Darius	1	49

- ▶ Each person's expected wellbeing: $\frac{1}{2} \cdot 49 + \frac{1}{2} \cdot 1 = 25$.

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- ▶ So the coin toss is better than (49,1) for sure, though it is no better in either state. Ex ante prioritarianism violates Statewise Dominance: Diamond's exit.

Ex Post Prioritarianism at Work

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Ex Post Prioritarianism at Work

Coin toss		
	Heads	Tails
Alex	49	1
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- ▶ Sum of $f(\text{wellbeing})$ if Heads: $\sqrt{49} + \sqrt{1} = 7 + 1 = 8$. If Tails: $\sqrt{1} + \sqrt{49} = 8$. Expectation: 8.
- ▶ (25, 25) for sure scores $5 + 5 = 10$ in every state.

The Two Versions at Work

	(49, 1) for sure	Coin toss	(25, 25) for sure
Ex ante	8	10	10
Ex post	8	8	10

With $f(w) = \sqrt{w}$.

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- ▶ Ex post: (25, 25) is better than the coin toss, though both give each person expected wellbeing 25. That goes against Ex Ante Pareto.
- ▶ So each version blocks a different move in the proof.

Ex Post Prioritarianism with One Person

	Heads	Tails
Sure thing	25	25
Gamble	49	1

Only one person exists. Her betterness is risk neutral about wellbeing.

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- ▶ Expected wellbeing 25 either way: equally good for her.
- ▶ Ex post prioritarianism: sure thing $\sqrt{25} = 5$; gamble $\frac{1}{2} \cdot 7 + \frac{1}{2} \cdot 1 = 4$. The sure thing is better.

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- ▶ So it violates Ex Ante Pareto even with nobody else around: no inequality, no unfairness.

Ex Ante and Ex Post Inequality

- ▶ *Ex post* inequality: in some state, people end up with different wellbeing.
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	Heads	Tails	Ex ante equal?	Ex post equal?
UNEQUAL	(10, 0)	(10, 0)	No	No
EQUAL CHANCES	(10, 0)	(0, 10)	Yes	No
EQUAL	(5, 5)	(5, 5)	Yes	Yes

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EQUAL	(5, 5)	(5, 5)	Yes	Yes

- ▶ EQUAL CHANCES: each person's expected wellbeing is 5, but whichever way the coin lands, one of them gets 10 and the other 0.

Ex Ante Equality, Any Ex Post Inequality

- Take any outcome, say (1, 2, 3). Spread it over three equally likely states, rotating who gets what (last week's lottery-equalisation):

	State 1	State 2	State 3
Alex	2	3	1
Darius	3	1	2
Philip	1	2	3

Each state has probability $\frac{1}{3}$.

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- Everyone's expected wellbeing is 2: ex ante equality.
- But every state is as unequal as (1, 2, 3). This works for any outcome: ex ante equality is compatible with any amount of ex post inequality.

Ex Post Equality Guarantees Ex Ante Equality

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- ▶ Not the other way round: EQUAL CHANCES is ex ante equal but unequal in every state.
- ▶ So guaranteed ex post equality is the more demanding kind (with a fixed population).

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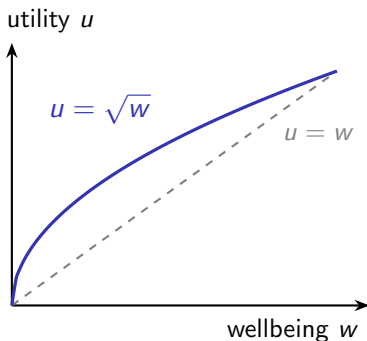
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- ▶ Ex post egalitarianism: EQUAL CHANCES guarantees an unequal outcome, so it's worse than EQUAL. That's the objection to Ex Ante Pareto.

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- ▶ Egalitarians face the same choice: inequality in people's chances, or inequality in outcomes?
- ▶ Ex ante egalitarianism: EQUAL CHANCES gives Alex and Darius the same chances (each has a 50–50 chance of 10 or 0), so it's better than UNEQUAL. That's Diamond's judgement, against Statewise Dominance.
- ▶ Ex post egalitarianism: EQUAL CHANCES guarantees an unequal outcome, so it's worse than EQUAL. That's the objection to Ex Ante Pareto.
- ▶ These aren't exclusive. An egalitarian can care about both ex ante and ex post inequality; so can a prioritarian.

Utility as a Function of Wellbeing

Two ways utility could curve against wellbeing.
The dashed line, $u = w$, is for comparison.

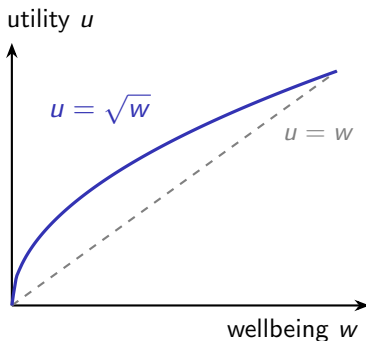


Concave:
risk averse about wellbeing

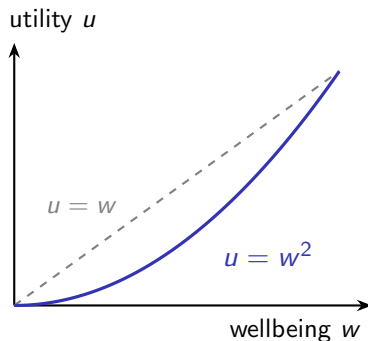
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Convex:
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Risk-Averse Betterness

- Now suppose instead that everyone's utility is $\sqrt{\text{wellbeing}}$.

Wellbeing	0	4	16
Utility	0	2	4

Risk-Averse Betterness

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Wellbeing	0	4	16
Utility	0	2	4

- ▶ A 50–50 chance of 16 or 0: expected utility $\frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 0 = 2$.

The Proof Again, with Risk-Averse Betterness

Everyone's utility is $\sqrt{\text{wellbeing}}$. Fair coin.

	Heads	Tails	Why
(16, 0) for sure	(16, 0)	(16, 0)	

Start from (16, 0) for sure.

The Proof Again, with Risk-Averse Betterness

Everyone's utility is $\sqrt{\text{wellbeing}}$. Fair coin.

	Heads	Tails	Why
(16, 0) for sure	(16, 0)	(16, 0)	Anonymity, Statewise Dominance
~ coin toss	(16, 0)	(0, 16)	

The first move is unchanged.

The Proof Again, with Risk-Averse Betterness

Everyone's utility is $\sqrt{\text{wellbeing}}$. Fair coin.

	Heads	Tails	Why
(16, 0) for sure	(16, 0)	(16, 0)	Anonymity, Statewise Dominance
~ coin toss	(16, 0)	(0, 16)	
~ (4, 4) for sure	(4, 4)	(4, 4)	Expected Utility, Ex Ante Pareto

Each person's 50–50 chance of 16 or 0 is as good for them as 4 for sure.

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~ (4, 4) for sure	(4, 4)	(4, 4)	Expected Utility, Ex Ante Pareto

By Transitivity: $(16, 0) \sim (4, 4)$.

The Same Steps, with Prioritarianism

Now everyone's utility is wellbeing, and we use ex post prioritarianism with $f(w) = \sqrt{w}$.

Priority-weighted total in each state:

	Heads	Tails	Expectation
(16, 0) for sure	$4 + 0 = 4$	$4 + 0 = 4$	4

The same three gambles as in the proof.

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All equally good: exactly the theorem's verdicts. Risk costs a person exactly what inequality costs overall.

Priority Equals Risk Aversion

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- ▶ So, given the premises, the priority given to the worse off must be exactly the degree of risk aversion about wellbeing.

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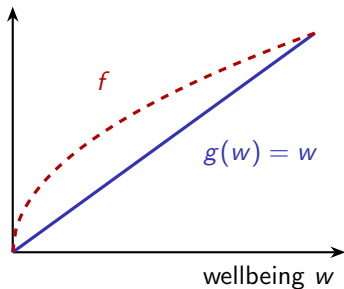
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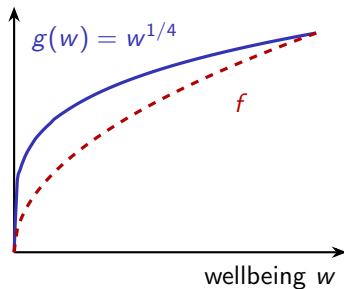
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- ▶ Her f could be more concave than g : priority beyond risk aversion.
- ▶ Or less concave: less priority than risk aversion, though still priority to the worse off in wellbeing.

Same Priority Weighting, Different Utilities

Solid: utility g . Dashed: priority weighting $f(w) = \sqrt{w}$.



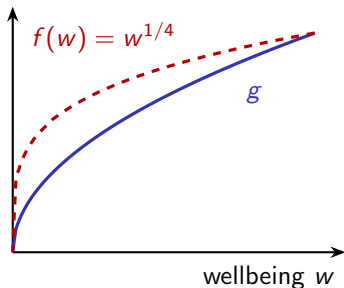
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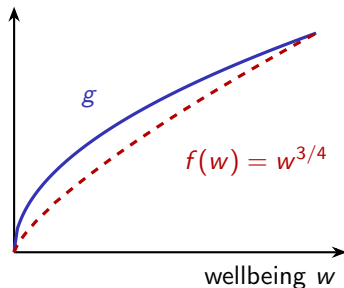
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Same Utility, Different Priority Weightings

Solid: utility $g(w) = \sqrt{w}$. Dashed: priority weighting f .



f more concave than g :
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- ▶ Example: $g(w) = \sqrt{w}$, so $w = u^2$.
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 - ▶ $f(w) = w^{3/4}$: $(f \circ g^{-1})(u) = (u^2)^{3/4} = u^{3/2}$: convex.
- ▶ So “priority to the worse off” means different things applied to wellbeing and applied to utilities.

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- ▶ So an egalitarian view that isn't separable over persons has to reject a premise.
- ▶ Last week's two objections were about how people fare relative to each other: Darius's chance compared with Alex's; one person sure to end up worse off than the other.

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